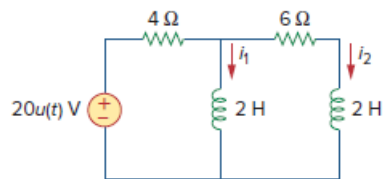


Problem

Obtain i_1 and i_2 for $t > 0$ in the circuit of Fig. 16.78.

Figure 16.78:



Step-by-step solution

Step 1 of 9

The Laplace transform of unit step function $u(t)$ is,

$$u(t) \leftrightarrow \frac{1}{s}$$

(Refer Figure 16.78 from textbook) draw the s-domain equivalent of the circuit.

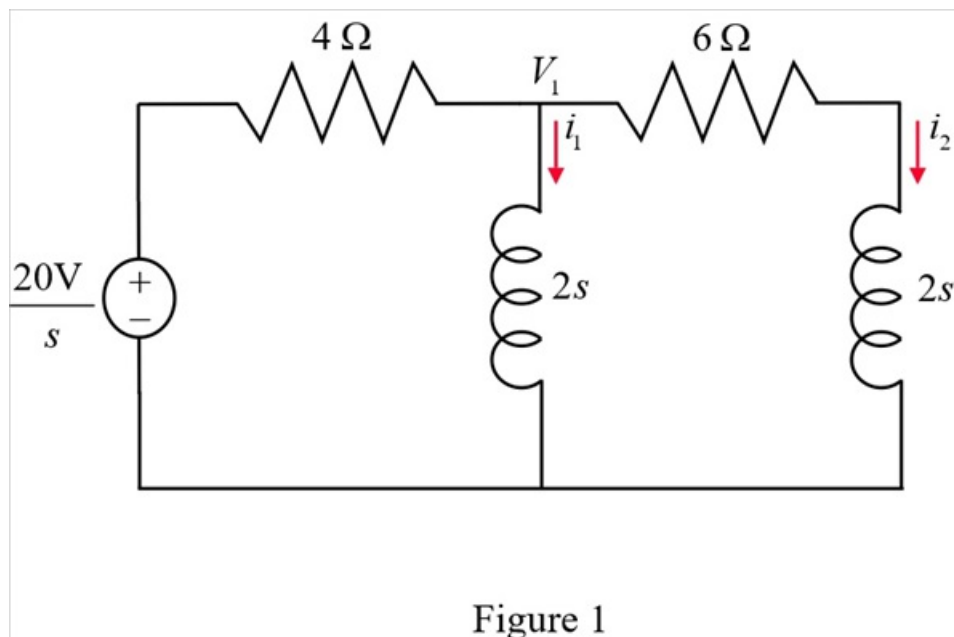


Figure 1

Step 2 of 9

Apply the nodal analysis in the above circuit.

$$\frac{V_1 - \left(\frac{20}{s}\right)}{4} + \frac{V_1}{2s} + \frac{V_1}{6 + 2s} = 0$$

$$\frac{(sV_1 - 20)(2)(6 + 2s) + 4V_1(6 + 2s) + 8sV_1}{8s(6 + 2s)} = 0$$

$$12sV_1 + 4s^2V_1 - 240 - 80s + 24V_1 + 8V_1s + 8V_1s = 0$$

$$4V_1[s^2 + 7s + 6] = 80s + 240$$

Hence, the node voltage V_1 is,

$$V_1 = \frac{20s + 60}{s^2 + 7s + 6}$$

Step 3 of 9

The current I_1 across the 2 H inductor is,

$$I_1 = \frac{V_1}{2s}$$

Substitute $\frac{20s + 60}{s^2 + 7s + 6}$ for V_1 .

$$I_1 = \frac{20s + 60}{2s(s^2 + 7s + 6)}$$

$$= \frac{10s + 30}{s(s^2 + 7s + 6)}$$

$$= \frac{10s + 30}{s(s + 1)(s + 6)} \dots\dots (1)$$

Step 4 of 9

Do the partial fraction of equation (1).

$$\frac{10s + 30}{s(s + 1)(s + 6)} = \frac{A}{s} + \frac{B}{(s + 1)} + \frac{C}{(s + 6)} \dots\dots (2)$$

$$10s + 30 = A(s + 1)(s + 6) + Bs(s + 6) + Cs(s + 1)$$

$$10s + 30 = s^2(A + B + C) + s(7A + 6B + C) + 6A$$

Compare the coefficient of s^2 , s and s^0 .

$$A = 5$$

$$B = -4$$

$$C = -1$$

Step 5 of 9

Substitute 5 for A , -4 for B and -1 for C in equation (2).

$$\frac{10s + 30}{s(s+1)(s+6)} = \frac{5}{s} - \frac{4}{(s+1)} - \frac{1}{(s+6)}$$

Rewrite the equation.

$$I_1 = \frac{5}{s} - \frac{4}{(s+1)} - \frac{1}{(s+6)}$$

Step 6 of 9

The shifting property of Laplace transform is,

$$\frac{1}{s+a} \leftrightarrow e^{-at}$$

Then, take the inverse Laplace transform of current I_1 .

$$i_1 = [5 - 4e^{-t} - 1e^{-6t}]u(t)$$

Hence, the current i_1 across the 2 H inductor is $\boxed{[5 - 4e^{-t} - 1e^{-6t}]u(t) \text{ A}}$.

Step 7 of 9

The current I_2 across the 2 H inductor is,

$$I_2 = \frac{V_1}{(2s+6)}$$

Substitute $\frac{20s+60}{s^2+7s+6}$ for V_1 .

$$\begin{aligned} I_2 &= \frac{20s+60}{(2s+6)(s^2+7s+6)} \\ &= \frac{10s+30}{(s+3)(s^2+7s+6)} \\ &= \frac{10}{(s+1)(s+6)} \dots\dots (3) \end{aligned}$$

Step 8 of 9

Do the partial fraction of equation (3).

$$\frac{10}{(s+1)(s+6)} = \frac{A}{(s+1)} + \frac{B}{(s+6)}$$

Substitute 2 for A and -2 for B .

$$\frac{10}{(s+1)(s+6)} = \frac{2}{(s+1)} - \frac{2}{(s+6)}$$

Rewrite the above equation.

$$I_2 = \frac{2}{(s+1)} - \frac{2}{(s+6)}$$

Step 9 of 9

The shifting property of Laplace transform is,

$$\frac{1}{s+a} \leftrightarrow e^{-at}$$

Then, take the inverse Laplace transform of current I_2 .

$$i_2 = (2e^{-t} - 2e^{-6t})u(t)$$

Hence, the current i_2 across the 2 H inductor is $\boxed{(2e^{-t} - 2e^{-6t})u(t) \text{ A}}$.